Cart lab: change driving force

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use controlled data to describe how net force affects acceleration.

SUCCESS CRITERIA - I can...

1. Change driving force while keeping total mass fixed.
2. Use data or a graph to compare accelerations.
3. Explain how resistance affects the ideal relationship.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A rover travels straight east at a steady 6 m/s. A second rover turns at the same steady speed. Which can have zero net force? Explain using velocity.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $a = F_{net}/M$; $F_{drive} = m_h g$.
Keep total moving mass M fixed. The driving weight equals net force only when resistance is negligible.

Worked reasoning: A 2 kg cart has net forces 2 and 6 N in two trials. Using a=F/m gives accelerations 1 and 3 m/s². Tripling net force triples acceleration.

Your turn: Draw a force-versus-acceleration graph for these trials. Explain the meaning of the slope.

Marzano check - circle: ① ② ③

03 Read a proportional pattern

Use the supplied model data for the same 2 kg system. What happens to acceleration when force doubles? Predict a at 8 N.

Net force (N)	a(m/s²)
2	1
4	2
6	3

Marzano check - circle: ① ② ③

04 Keep mass constant

A student adds a new mass from the supply box to the hanger. What two quantities change? Describe a better method for this experiment.

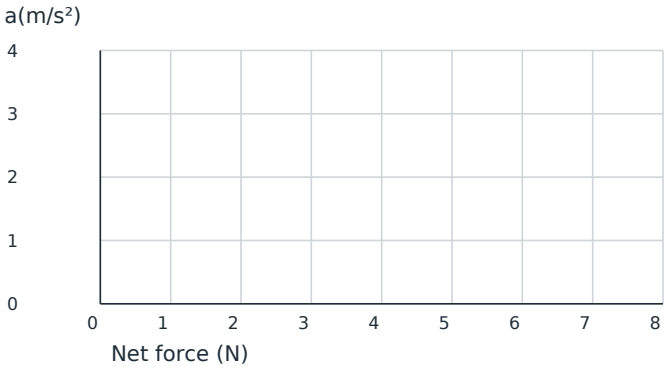
Marzano check - circle: ① ② ③

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05 Graph the pattern

Plot the model data from 3. Label both axes and draw a trend line. If you collect your own data, put those in a separate table and label them measured.



Marzano check - circle: 1 2 3

06 Choose a method

With a fixed net force, a 3 kg system accelerates at 10 m/s². Predict acceleration for a 6 kg system. State a control needed for this comparison.

Marzano check - circle: 1 2 3

07 Exit check

A fixed-mass system accelerates at 2 m/s² under 4 N net force. Predict its acceleration under 6 N net force. Explain with a comparison.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Design three net-force settings that give accelerations 1, 2, 3 m/s² for a 3 kg cart. Explain what must stay fixed.

Marzano check - circle: 1 2 3